

Java Identifiers Mini-Book

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1. What are Identifiers in Java?

In Java, an identifier is a name used to identify classes, methods, variables, arrays, or labels.

Real-world example:

Identifiers are like name tags at a party. Each person has a unique tag for recognition, just like variables or classes in a program.

2. Where are Identifiers Used?

- Class names: `class Student {}`
 - Method names: `void printDetails() {}`
 - Variable names: `int age = 25;`
 - Label names: `L1: while(true) {}`
 - Array names: `int[] marks = new int[5];`
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3. Rules for Naming Identifiers

Allowed Characters:

- A-Z, a-z
- 0-9 (but not at the beginning)
- `_` (underscore)
- `$` (dollar sign)

Not Allowed:

- Special symbols like `#`, `@`, `!`
- `.`

Java reserved keywords (e.g., `int`, `if`, `class`)

Practice:

- `Total number`
 - `123total`
 - `total123`
 - `Total#`
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4. Are Identifiers Case Sensitive?

⇒ Yes, Java is case-sensitive. Uppercase and lowercase letters are treated as distinct.

Example:

```
int number = 10;  
int Number = 20;  
int NUMBER = 30;
```

All three are distinct identifiers.

5. Can Identifiers Be Very Long?

⇒ There is no technical limit to identifier length, but long names reduce readability.

Bad:

```
int theTotalNumberOfItemsPurchasedInStore = 500;
```

Good:

```
int itemCount = 500;
```

6. Can You Use Java Keywords as Identifiers?

⇒ No, Java reserved keywords (e.g., `if`, `class`, `int`) cannot be used as identifiers.

Example:

```
int class = 10; // Invalid: Compile-time error
int myClass = 10; // Valid
```

7. Can You Use Java Class Names as Identifiers?

⇒ Yes, it's syntactically valid but discouraged as it reduces code clarity.

Example:

```
int String = 888;
int Runnable =
99;
System.out.println(String); // Prints 888
System.out.println(Runnable); // Prints 99
```

8. Valid vs Invalid Identifiers

Valid Identifiers:

- Total_number
- Total123
- Ca\$h
- \$a\$b\$c
- java2Shape
- Integer

Invalid Identifiers:

- Total#
 - 123total
 - all@hands
 - int (keyword)
-

9. Interview Questions & Answers

Q1. What are identifiers in Java?

A: Identifiers are names used for variables, methods, classes, arrays, and labels.

Q2. What characters are allowed in identifiers?

A: A-Z, a-z, 0-9 (not at the start), `_`, and `$`.

Q3. Are identifiers case-sensitive?

A: Yes, `data`, `Data`, and `DATA` are treated as different identifiers.

Q4. Can we use reserved words as identifiers?

A: No, reserved words like `int`, `if`, or `class` cannot be used.

Q5. Can we use predefined class names as identifiers?

A: Yes, but it's not recommended as it can reduce readability.

Q6. Can an identifier start with a digit?

A: No, identifiers cannot begin with a digit.

Q7. How to choose good identifier names?

A: Use concise, descriptive names like `totalMarks` instead of `t` or `theTotalMarksOfSubjectInExamFor2024`.

10. Homework (Try it Yourself!)

1. Create a class with valid names for class, method, and variable.
 2. Try declaring a variable starting with a digit.
 3. Use `class` as a variable name.
 4. Use both `name` and `Name` as variable names.
 5. List 5 valid and 5 invalid identifiers.
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11. Homework Solutions

1. Valid class, method, and variable:

```
class Hello {  
    void greet() {  
        int age = 25;  
    }  
}
```

2. Variable starting with a digit:

```
int 1count = 10; // Invalid: Compile-time error
```

3. Using `class` as a variable name:

```
int class = 10; // Invalid: Compile-time error
```

4. Using `name` and `Name`:

```
int name = 10; int Name = 20; // Both are  
valid and distinct
```

5. Valid and Invalid Identifiers:

- **Valid:** `user1`, `_value`, `$count`, `speed100`, `Main`
- **Invalid:** `1stValue`, `class`, `@home`, `high-score`, `float`

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